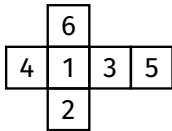
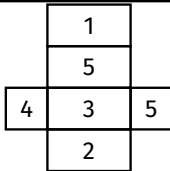


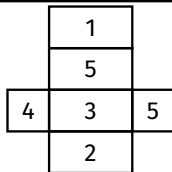
Push Tasks - Nets and SA



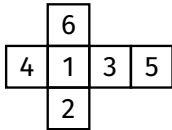
1. Invalid arrangements?
e.g. 4 in a row with 2 on
same side



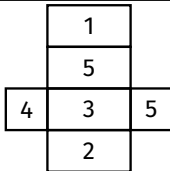
3. 3 pairs face areas?
15, 10, 6 each x 2



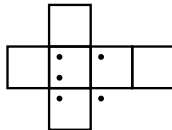
5. Max SA for $V=96$?
 $1 \times 1 \times 96 = 386 \text{ cm}^2$



2. Why some fail?
Overlap when folded

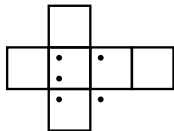


4. Min SA for $V=96$?



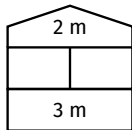
6. Opposite sums to 7?
Standard: 1-6 2-5 3-4

Push Tasks - Nets and SA



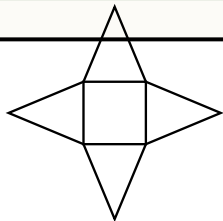
7. Different die net?

Any valid cube net

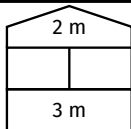


9. Different net?

Swap triangle positions

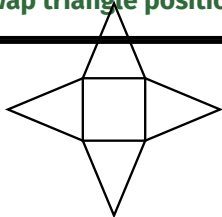


11. Pyramid SA: base 25, tri 15 each?

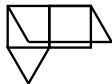


8. Triangles isosceles?

Two equal sides by



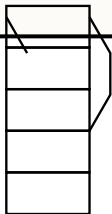
10. Triangles congruent?



12. Right triangle prism net?

Rectangles + 2 triangles

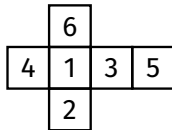
Push Tasks - Nets and SA



13. Hex prism SA side 2,
len 5?



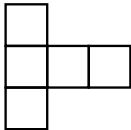
15. Why 4 triangles?
4 triangular faces



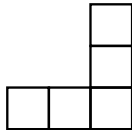
17. Shortest surface path?
Unfold net, draw line



14. Tetrahedron nets?
2 distinct nets

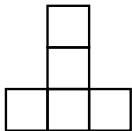


16. $V=12$ no top SA?
Depends on dimensions



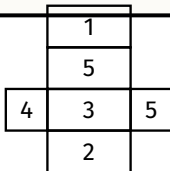
18. L as cube net?
Only with 6th square

Push Tasks - Nets and SA



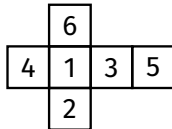
19. 2 non-box pentominoes?

I and X pentominoes



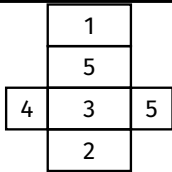
21. 4x6x4 min paper proof?

Closest to cube

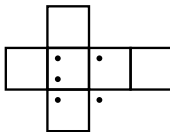


23. Shortest path?

Draw net, straight line

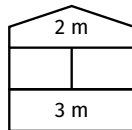


20. 1x1x96 max paper?



22. Min colours?

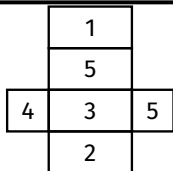
3 (opposite faces same)



24. Ribbon on faces?

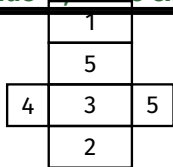
3 8 m on faces

Push Tasks - Nets and SA

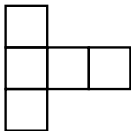


25. SA=150, side then V?

Side=5, $V=125 \text{ cm}^3$



26. $8 \times 3 \times 2$ wrap cost
 $\$0.05/\text{cm}^2$?



27. Open-top max
volume?

Depends on card size